

Lab # 1

Programming Assignment: Sequence Alignment Using Edit Distance Algorithms

Objective

The objective of this assignment is to:

- Understand and implement global and local alignment techniques using dynamic programming.
- Explore the principles of minimum edit distances in the context of DNA sequence alignment.
- Analyze the impact of scoring parameters (match, mismatch, gap penalties) on the alignment results.
- Gain practical experience with biological data processing and algorithm evaluation.

Background Knowledge Required

- Dynamic programming fundamentals.
- Understanding of edit distance and string similarity metrics.
- Basic bioinformatics concepts such as nucleotide sequences and sequence alignment.
- Python programming proficiency, including numpy and matplotlib.

Dataset

Use the Homo sapiens mitochondrial DNA sequences or any dataset from:

- NCBI GenBank: <https://www.ncbi.nlm.nih.gov/genbank/>
- Alternatively, use sample data from the Biopython package:

```
from Bio import SeqIO  
sequence = list(SeqIO.parse("path/to/fasta_file.fasta", "fasta"))
```

Provide students with (see last section below):

- A subset of 10 pairs of DNA sequences (~100–300 bp each) for alignment tasks.

Assignment Tasks

Part 1: Implement Alignment Algorithms

1. Implement Needleman-Wunsch (Global Alignment) from scratch using dynamic programming.
2. Implement Smith-Waterman (Local Alignment) similarly from scratch.

Scoring:

- Match score: +2
- Mismatch penalty: -1
- Gap penalty: -2

Part 2: Visualization

- Visualize the alignment matrices (score + traceback) using heatmaps (e.g., matplotlib or seaborn).
- Display aligned sequences with matches and mismatches clearly indicated.

Part 3: Parameter Sensitivity

- Modify match/mismatch/gap parameters and re-run alignment on 3 selected pairs.
- Report the changes in alignment results and explain.

Part 4: Bonus (Optional)

- Compare your implementations with Biopython's pairwise2 module and report time and result differences.

Python Libraries to Be Used

- numpy – for matrix operations
- matplotlib / seaborn – for visualizations
- Biopython – for loading and comparing biological sequence data
- time – for benchmarking

Deliverables

- alignment.py: Script with both implementations.
- results/: Folder containing alignment outputs (raw scores and alignments).
- report.pdf: A 3–5 page report including:
 - Description of algorithms
 - Discussion of alignment outputs (including visualizations)
 - Analysis of parameter effects
 - Comparison with Biopython results (if completed)
- README.md: Brief guide to running the code.

Evaluation Criteria (Total: 100 pts)

Component	Points
Correctness of Needleman-Wunsch algorithm	20
Correctness of Smith-Waterman algorithm	20
Matrix construction and traceback logic	20
Visualization quality and clarity	10
Report quality and insights	20
Code structure, readability, documentation	10
Bonus: Comparison with Biopython (optional)	+5
Bonus: Parameter sensitivity analysis	+5

Pedagogical Rationale

This assignment anchors the abstract concept of edit distances in a meaningful biological context, making it more engaging. The requirement to implement both global and local alignment emphasizes the subtle but important differences in alignment strategies. Visualizing the matrices and alignment paths deepens understanding, while parameter tuning fosters critical thinking about algorithm behavior.

10 Pairs of DNA sequence

Pair 1

Sequence 1:

GACTTACGCGCCGTAGCACTTCTGTGATAGCTGCGAGGCGTATTGCTACTTGTACGAGATAGGGTCGACTTT
TCGGAGTCGACAGACACTACGATACT

Sequence 2 (Reverse Complement):

AGTATCGTAGTGTCTGTGCGACTCCGAAAAGTCGACCCTATCTCGTACAAGTAGCAATACGCCTCGCAGCTAT
CACAGAAGTGCTACGGCGCGTAAGTC

Pair 2

Sequence 1:

GTAGTCATTACCCATAGGCGTTTGCTGACTATCGGGAGAGGTACGTACGTGATACGACTGTGCGATCGCTCGT
AGATCAAGATACTCAGCGTGCTGTC

Sequence 2 (Reverse Complement):

GACAGCACGCTGAGTATCTTGATCTACGATCGACAGTCGTATCACGTACGTACCTCTCCCGATAGTCAGCAA
ACGCCTATGGGTAATGACTAC

Pair 3

Sequence 1:

CTGCATACCGACAAACGTGTGCATGCATGCATGCATGCATGCATGCATGCATGCATGCATGCATGCACATGC
ATACGTGCGTGTCTACGTACGT

Sequence 2 (Reverse Complement):

ACGTACGTCTACGCACGTATGCATGTGCATGCATGCATGCATGCATGCATGCATGCATGCATGCATGCATGC
ACACGTTTGTGCGTATGCAG

Pair 4

Sequence 1:

[illegible]

Pair 9

Sequence 1:

[illegible]

Sequence 2 (Reverse Complement):

[illegible]

Pair 10

Sequence 1:

[illegible]

Sequence 2 (Reverse Complement):

CTAGCTAGCTAGCTAGCTAGCTAGCTAGCTAGCTAGCTAGCTAGCTAGCTAGCTAGCTATCGCT
ATCGCG

Due Date

Monday 14, September 2026, 11:59pm. Teams are going to be asked to present the lab in front of the class. These teams are going to be randomly selected, so all teams must be ready. The presentation should be done directly from the jupyter notebook.